

## **Practical Project:**

### **Title: Comparative Study between Machine Learning and Deep Learning for Image Classification**

#### **Task Description:**

Each group is required to perform an image classification task using **one dataset only** selected from either:

- **scikit-learn datasets**
- **Kaggle image datasets**

The objective is to conduct a comparative study between **Machine Learning (ML)** and **Deep Learning (DL)** approaches, including **transfer learning using pretrained CNN models**.

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#### **Requirements:**

Each group must:

- Select **one approved image dataset**
  - Implement at least:
    - **Two Machine Learning model**
    - **Two Deep Learning pretrained CNN models**
  - Train, validate, and test all models
  - Perform a detailed comparison between all approaches
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#### **Machine Learning Approach:**

- Apply feature extraction then use classical ML algorithms such as: Support Vector Machine (SVM) or K-Nearest Neighbors (KNN) ,.....etc
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#### **Deep Learning Approach (CNN):**

- Use **two different pretrained models** (e.g., ResNet, VGG, MobileNet)
  - Apply **transfer learning and fine-tuning**
  - Use **data augmentation techniques**
  - Train models on the selected dataset
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## **Evaluation and Analysis:**

Students are required to calculate:

- Accuracy
- Precision
- Recall
- F1-score

### ◆ **Visualizations:**

- Confusion Matrix
- Training vs Validation curves

### ◆ **Comparison:**

- ML vs DL performance
- Comparison between pretrained CNN models

### ◆ **Analysis:**

- Training complexity
  - Feature engineering vs automatic feature learning
  - Effect of dataset characteristics
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## **Final Deliverables:**

### **1. Research Paper (Mandatory)**

Must include:

- Introduction
- Dataset Description
- Methodology (ML + DL models)
- Experiments and Results
- Discussion
- Conclusion
- References

### **2. Source Code**

Well-documented and organized

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## **Oral Discussion:**

Students will be evaluated on:

- Understanding of ML and DL concepts
- Ability to explain results
- Justification of model choices
- Answering questions